Introduction to Biophysics II: Quantum physics & biology

Lecture 23:

Sense of smell:
Smelling quantum vibrations

Olfaction*:* the action or capacity of smelling; the sense of smell

Task: Detect types of molecules *(odorants)* based on different physical and chemical properties.

Reading:
Quantum effects in biology,
M. Moshedi, Y. Omar, GS Engel, M, Plenio
Chapter 12

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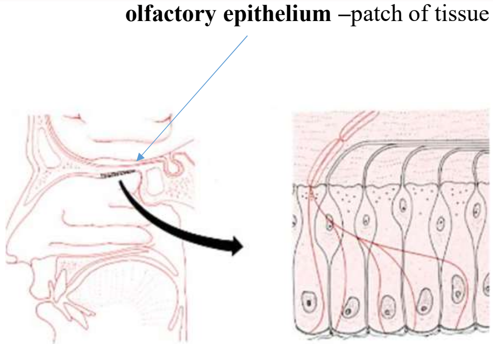
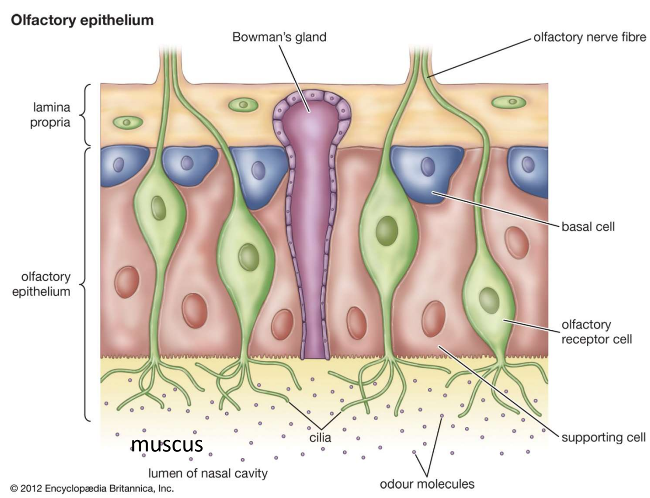
Physiology of olfaction

olfactory epithelium –patch of tissue the size of a postage stamp:





Contains:
- sensory neurons, each with a primary cilium
- supporting cells between them
- basal cells that divide regularly, producing a fresh crop of sensory neurons to replace those that die (and providing an exception to the usual rule that neurons seldom are replaced).

https://www.biology-pages.info/O/Olfaction.html https://www.britannica.com/science/olfactory-epithelium

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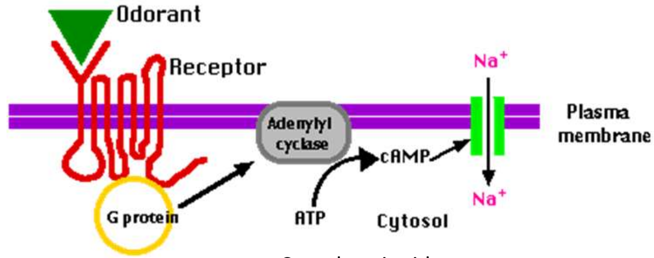
Physiology of olfaction

Odorant receptors:



Cytoplasmic side

<u>The sequence of events</u>:
- Odorant molecules dissolve in the mucus and bind to receptors on the cilia
- Binding of the odorant activates a G protein
- G protein activates adenylyl cyclase that catalyzes the conversion of ATP to cyclic AMP (cAMP) (see EXTRA section)
- cAMP opens Na^+ channels and initiates an *action potential*
- That is conducted along the olfactory nerve to the brain

Human genome: ~1000 genes encoding different odor receptors.
Highly nonlinear: can detect molecules of different types simultaneously, even if concentrations are 1000x different

https://www.biology-pages.info/O/Olfaction.html

Proposed mechanisms of olfaction

State of research:
- Only primary structures - sequences of amino acids – are known; no X-ray structures
- Detailed mechanism is unknown, despite century-long research.
- Theories are still to be proven in an unambiguous way
- No theory has shown predictive power

What is known:
- Odorant molecules should be volatile
- Must contain fewer than 16 carbons (i.e., mass <240 Daltons, <50 atoms)
- No two odorants smell exactly the same
- But: enantiomer pairs often smell the same

Possible mechanism:
- *shape-based molecular recognition* governed olfaction, as it does much of the rest of biology (Pauling, 1946)
  Problem – enantiomers will have a “different” odor.
- *Vibrational recognition*: (Dyson, 1938**):** The nose is a “vibrational” spectroscope, which detects a molecule via its vibrational spectrum.
  This theory is attractive: one sensor could recognize even new molecules via vibrational signature without inventing new receptors.
  Theory survived for 4 decades with no molecular mechanism proposed till recently…

Experimental observations

Some experimental facts:

1. Humans can easily detect the presence of certain functional groups in an odorant, whatever their molecular context.
   Example: "-SH" group, attached to any molecule, smells sulfurous. "Shape recognition" cannot discriminate perfectly between thiols and alcohols at all concentrations – it is a very challenging task.

2. The majority of enantiomer pairs have very similar, or in many cases identical, smells. "Shape recognition" based on chiral proteins will sense them differently. Known shape-recognition receptors bind one or the other.

3. Replacement of carbon by silicon, germanium, or tin (same group in the periodic table) leads to large changes in odor. These substitutions will not change the shape or chemical properties of the odorant.

4. Isotope effect: Replacement of hydrogen by deuterium is perceived as a different odor by fruit flies and humans.
   Some reservations: hard to make pure isotopomers, and olfaction can detect two compounds simultaneously, even when concentrations are 1000x different.
   However, some experiments are convincing (Franco et al., 2011)
   - Trained fruit flies to avoid either hydrogen or deuterium isotopomers.
   - Flies that avoided –C-H group also avoided -C≡N group – both have similar stretch vibration frequencies

5. Newly synthesized chemical odorants can be smelled *immediately*; no new receptors are needed

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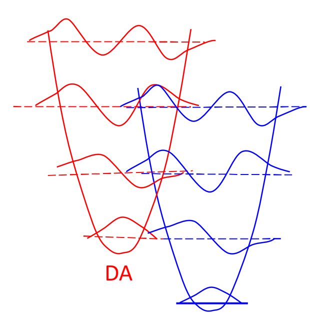
Vibration role in electron transfer

Proposed mechanism (Turin 1996):

Basis:
- molecular vibrations are *quantized*
- An electron can tunnel over barriers

Lecture 16-18

$$k_{ET} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \times$$
$$\times \sum_{n,m}\left[\frac{\exp\left(-E_{D,n}/kT\right)}{Z_1}\left|\left\langle \chi_{A,m} \mid \chi_{D,n}\right\rangle\right|^2 \times \boxed{\delta\left(E_{A,m}-E_{D,n}+\Delta E_{DA}\right)}\right]$$
$$\times\left\langle \sigma_A \mid \sigma_D\right\rangle$$



An efficient electron transfer energy conservation requires a vibration, which can efficiently "absorb" extra energy during electron transfer.

An odorant molecule can provide such an *intramolecular* vibration tuned to the energy difference between DA and D^+A^- greatly enhancing electron transfer probability

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Vibrational recognition

Proposed mechanism (Turin 1996)**:** vibration-frequency detection by electron transfer

- Electron transfer must be between sites with a well-defined energy difference

If electron transfer *alone* determines odor (no shape dependence):
- Isotope dependence of scent (observed)
- Enantiomers will have the same scent (observed)
- Link between infrared spectra and odor (infrared spectra probe absorption of vibrational transitions) (verified for hydrogen sulfide and decaborane (Brookes et al., 2007)

For this mechanism to be viable:
- Tunneling rates, vibrational parameters, and Franck-Condon factors must be suitable
- Must work at room temperature
- Nature often uses the same mechanism for various receptors – perhaps some other small molecule receptors work in the same way (hormones, steroids, neurotransmitters)

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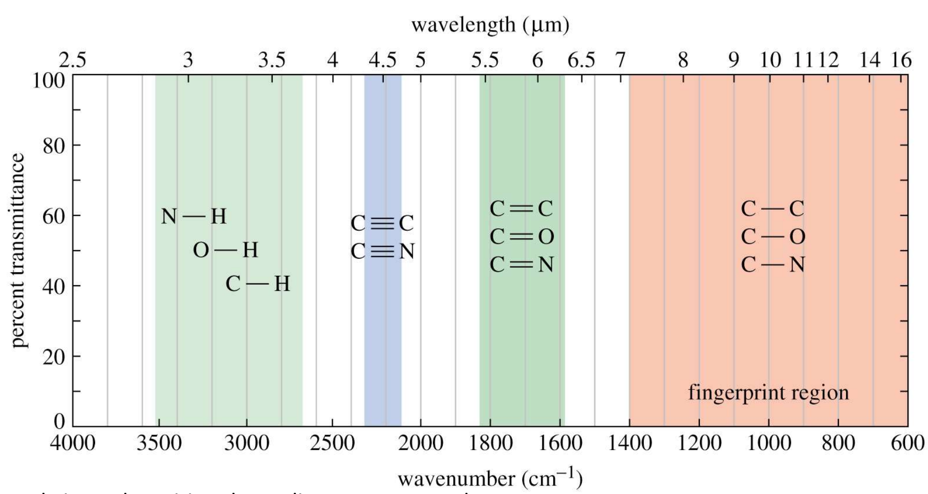
Molecular vibrations: energy scale

Intramolecular vibrations and room temperature:

$kT_{RT} \approx 200\ cm^{-1}$ (25 meV)

Stretch frequencies:→



"Fingerprint region" varies greatly in peak position depending on compound

https://wps.prenhall.com/wps/media/objects/340/348272/wade_ch12.html

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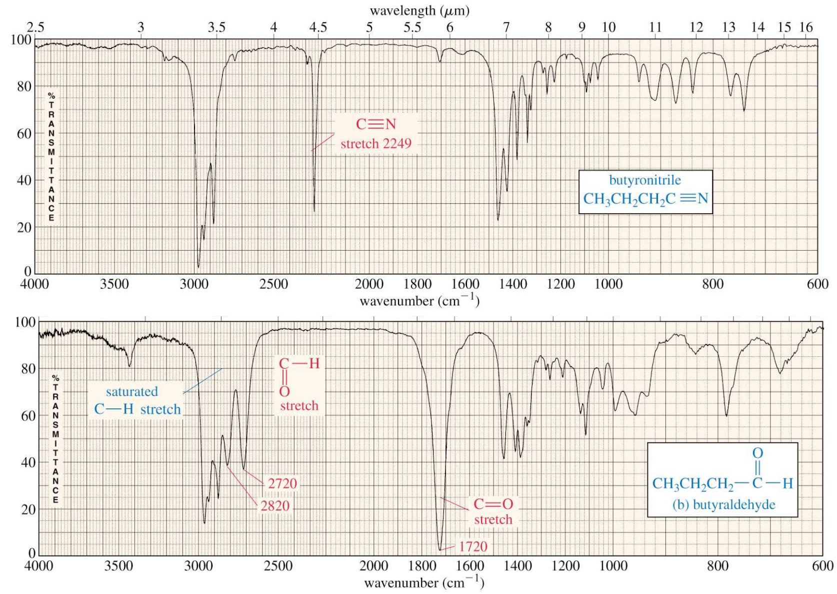
Molecular vibrations: examples

Example:



https://wps.prenhall.com/wps/media/objects/340/348272/wade_ch12.html

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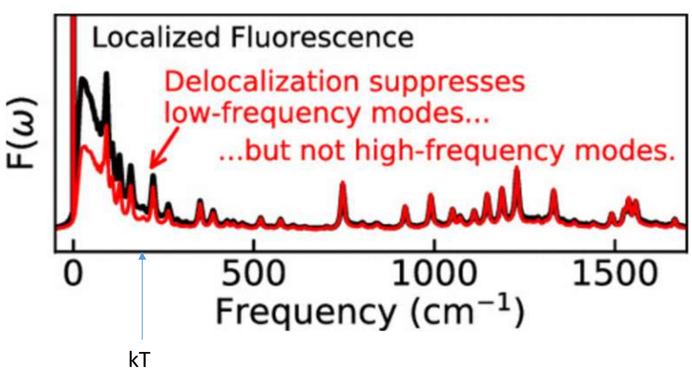
Intra molecular vibrations and bath: example

Lecture 01-05

Molecular vibrations:

Example: chlorophyll *a*



Mike Reppert: . *Phys. Chem. B* 2020, 124, 45, 10024–10033

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Proposed sequence of odor detection

Odor recognition: The event sequence:

(i) odorant molecules begin in the atmosphere;
(ii) they then enter the nasal cavity.
(iii) they bind at some ensemble of receptors, leading to
(iv) selective actuation of one or more receptors, generating signals that are amplified and transmitted to the brain,
(v) where they are analyzed.
(vi) Possibly, some analysis may occur before the brain is reached. The system as a whole imposes some constraints, but it is a process that we assess thoroughly and propose must be quantal.

What we know:
- Primary structure of receptor protein (sequence of amino acids) but not 3D structure
- Olfactory receptors are most probably hydrophobic and chiral (as other receptors), expected 7 transmembrane α-helices
- Expect significant thermal fluctuations enabling change of shape upon binding of odorant molecule
- Overall process of recognition occurs in milliseconds (slow compared to electron transfer in photosynthesis)

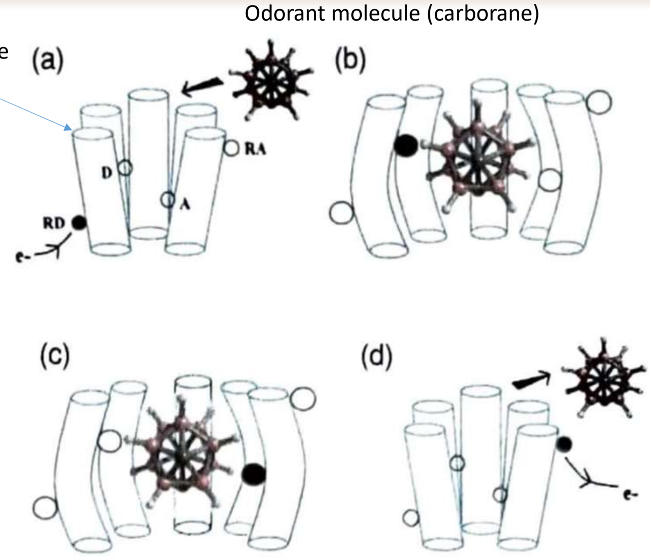
Proposed mechanisms of olfaction

Proposed mechanism:

a) An electron diffuses to the RD position;
b) Odorant docks causing configurational change of the protein;
Electron transfers (tunnels) to D and spends most time there
c) Electron tunnels D→A exciting certain intramolecular vibration of odorant
d) Odorant is expelled from the cavity and electron transfers to the signaling site RA, initiating G-protein release

Note: for electron transfer to be efficient, the intramolecular vibration of the odorant must match the energy difference between D^-A and DA^-.

(Note: 5 helices are shown out of 7)



Proposed mechanisms of olfaction

Requirements:

Need energy input to produce initial state - get electron onto RD site.

- Mechanism not known. May include some reducing/oxidizing species “X” in a cell. Analogy: plastoquinol or ferredoxin transfers electrons in photosynthesis and respiration

Need electron (hole?) carriers to diffuse to site RD:

- Need time:

  $\tau_X = r\eta/(2n_X kT)$ (Stokes-Einstein equation)

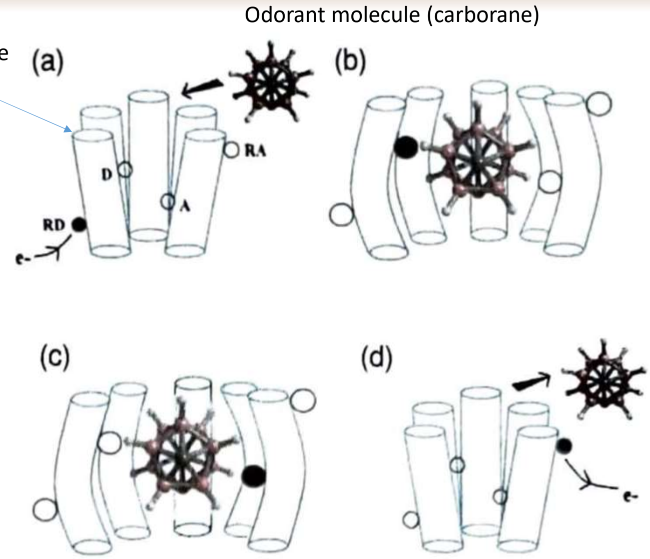
  n_X - concentration

  η - viscosity of water (membrane?)

For typical values n_X~1-100μM τ_X~0.1–1 ms (in H_2O)

Need electron transfer to occur:

- *Time needed is up to milliseconds (Marcus) for typical proteins*



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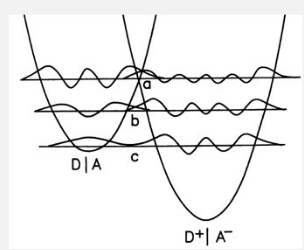
Huang-Rhys and Franck-Condon factors

Vibration assisted electron transfer **Lecture 10, 16-18**

$$k_{ET} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \left(4\pi\lambda_{bath}kT\right)^{-1/2} \sum_{n,m} \left[\begin{array}{l} \dfrac{\exp\left(-E_{D,n}/kT\right)}{Z_1} \left|\left\langle \chi_{A,m} \mid \chi_{D,n} \right\rangle\right|^2 \times \\ \times \exp\left(-\dfrac{\left[\lambda_{bath} + E_{A,m} - E_{D,n} + \Delta E_{DA}\right]^2}{4\lambda_{bath}kT} \right) \end{array} \right]$$



Franck-Condon factor $\boxed{\left|\langle \chi_m | \chi_0 \rangle\right|^2 = \dfrac{S^m \exp(-S)}{m!}}$

Huang-Rhys factor: $S = \dfrac{\lambda_{internal}}{h\nu}$

Can estimate Huang-Rhys factors computationally…

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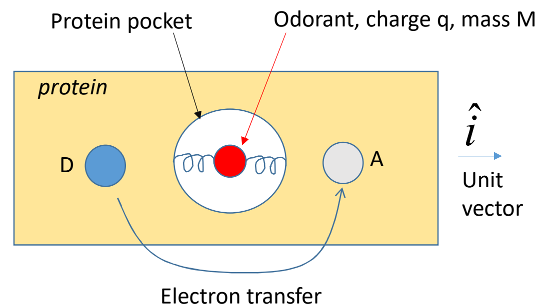
Mechanistic model

Simple model:

Odoroant (its atom): charge q, mass M is held in the cavity with spring of stiffness k: resonance frequency ω: $\omega^2=k/M$.

Electron transfers (abruptly) from D to A: electric field E changes and kicks the odorant charge.



Unbalanced Force: $F=\Delta E q$

Shift from new equilibrium: $\Delta x = F/k$

Reorganization energy: $\lambda = \frac{k\Delta x^2}{2}=\frac{kF^2}{2k^2}=\frac{\Delta E^2 q^2}{2k}$

Huang-Rhys factor: $S=\frac{\lambda}{\hbar\omega}=\frac{\Delta E^2 q^2}{2k\hbar\omega}=\frac{\Delta E^2 q^2}{2M\hbar\omega^3}$ ← $\omega^2=k/M \rightarrow k = M\omega^2$

Electric field change: $\Delta E=\frac{e}{4\pi\varepsilon}\left(\frac{\vec{R}_D}{R_D^3}-\frac{\vec{R}_A}{R_A^3}\right)=\frac{e}{4\pi\varepsilon}\left(\frac{1}{R_D^2}-\frac{1}{R_A^2}\right)\hat{i}$

R_D – vector from odorant to D
R_A – vector from odorant to A
(both along i in this example)

Recall FC: $|\langle\chi_m|\chi_0\rangle|^2=\frac{S^m\exp(-S)}{m!}$

- Relevant Huang-Rhys (and FC) falls off with distance (R^{-4}) to A and D
- Frequency of vibration should match ω for energy conservation

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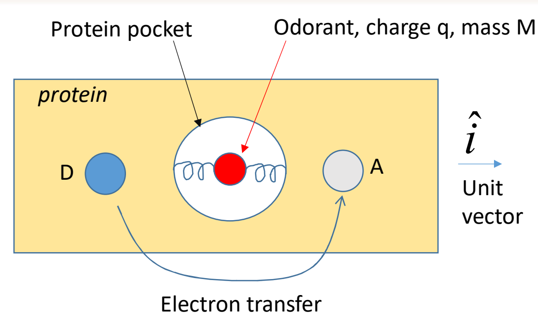
Mechanistic model

Simple model:

$S=\frac{\Delta E^2 q^2}{2M\hbar\omega^3}$ $\qquad$ $\Delta E=\frac{e}{4\pi\varepsilon}\left(\frac{\vec{R}_D}{R_D^3}-\frac{\vec{R}_A}{R_A^3}\right)\hat{i}$

$|\langle\chi_m|\chi_0\rangle|^2=\frac{S^m\exp(-S)}{m!}$



Notes:
- Huang-Rhys (and FC) falls off with distance ~R^{-4} to A and D
- Frequency of vibration should match $m\omega$ for energy conservation
- The ratios of zero-phonon, one-phonon, etc. transitions are fixed
- All significant oscillations of odorant are in the ground state (>>kT)

Also important – electron transfer rate, ratio of elastic rate (no energy loss to vibration, $m=0$), and inelastic ($m=1$ vibration excited)

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Estimating vibrational effect magnitude

RECALL: Electron transfer rate in approximation where solvent and intramolecular vibrations are split (Lecture 15):

$$k_{ET} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \frac{1}{\sqrt{4\pi\lambda_{bath}kT}} \sum_m \left[\sigma_m \times \exp\left(-\frac{\left[\lambda_{bath} + \Delta E_{DA} + mh\nu\right]^2}{4\lambda_{bath}kT}\right)\right] \qquad \sigma_m = \frac{e^{-S}S^m}{m!}$$

(Franck-Condon series)

(high-frequency internal vibrations replaced with one at frequency ν)

Important: inelastic electron transfer rate (m=1, excite odorant vibration) must be much larger than elastic *(m=0)* rate :

$$k_{ET,m} = \frac{1}{\tau_{ET,m}} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \frac{\sigma_m}{\sqrt{4\pi\lambda_{bath}kT}} \exp\left(-\frac{\left[\lambda_{bath} + \Delta E_{DA} + m\hbar\omega\right]^2}{4\lambda_{bath}kT}\right)$$

Typical parameters:

$\hbar\omega$	S	λ	$\lvert H_{AD}\rvert$
200 meV	0.1	30 meV	1 meV

Brookes et al. (2007),

$\tau_{ET,0}$ = 87 ns
$\tau_{ET,1}$ = 0.15 ns

→ ***Electron transfer rate can be enhanced >100 times in the presence of odorant***

Requirement: The reorganizational energy of the protein *must be* small for good selectivity. Partly supported by the hydrophobic nature of the pocket (mostly non-polar amino acids).
Problem: While intramolecular vibrations of odorant are easy to model, the reorganizational energy of the environment (bath) is not – we do not even have the structure of these proteins.

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Estimating vibrational effect magnitude

Estimate coupling energy $|H_{AD}|$

Assume: odorant M serves as a "bridge" between A and D, i.e., electron transfer A→D goes via "virtual" M (recall lecture 16-18):

$$H_{AD(el)} = V_{DM}\frac{1}{(E_M - E_A)}V_{MA} = \frac{V^2}{(E_M - E_A)}$$

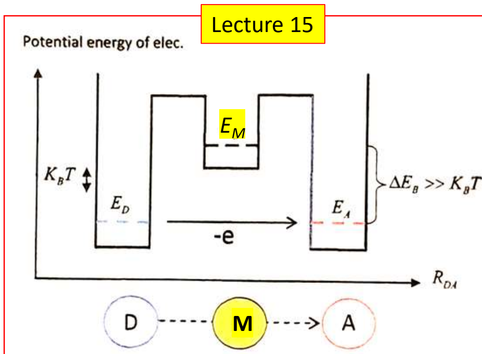
(assume couplings V_{DM}=V_{MA}=V)

If E_M corresponds to the LUMO orbital of M and E_A corresponds to the HOMO orbital at the time of transition, then the difference can be as large as $(E_M - E_A)$ ~ 10 eV.

Assume bonds between M and A and D are comparable to hydrogen bonds (0.05 – 0.5 eV) → take V ~ 0.1 eV

→ $|H_{AD}|$ ~ 1 meV ← as used on previous slide, makes sense.

Lecture 15



$$Path_i(D, B_{i1}, B_{i2}, ..., B_{iN}, A) = \\ = V_{D,B_{i1}}\frac{1}{\Delta E_{i1}}V_{B_{i1},B_{i2}}\frac{1}{\Delta E_{i2}}V_{B_{i2},B_{i3}}\cdots\frac{1}{\Delta E_{iN}}V_{B_{iN},A}$$

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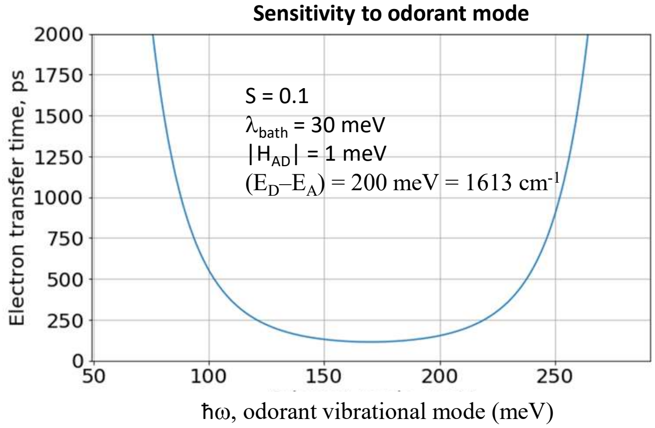
Estimating vibrational effect magnitude

ET Transfer modeling: $k_{ET,m} = \frac{1}{\tau_{ET,m}} = \frac{2\pi}{\hbar} \left| H_{AD(el)} \right|^2 \frac{\sigma_m}{\sqrt{4\pi\lambda_{bath} kT}} \exp\left(-\frac{\left[\lambda_{bath} + \Delta E_{DA} + m\hbar\omega \right]^2}{4\lambda_{bath} kT} \right)$ $\sigma_m = \frac{e^{-S} S^m}{m!}$

Sensitivity to odorant mode




```
import matplotlib.pyplot as plt
from matplotlib import cm
import numpy as np
import math as math

def fc(S=0.1,m=1):
    return math.exp(-S)*(S**m)/math.factorial(m)

def rate(Had=0.001,S=0.1,m=1,lmd=0.03,dE=0.2,vib=0.2):
    hbar = 6.58e-16     # eV*s
    kT = 0.0259          # eV at RT (300K)
    k = 2*math.pi/hbar * (Had**2) * fc(S,m) / math.sqrt(4*math.pi*lmd*kT) * math.exp(-(dE-m*vib-
lmd)**2/(4*kT*lmd))
    return k

y = []
y2 = []
x = []

for z in np.arange(.06,0.28,0.001):
    k = rate(Had=0.001,S=0.1,m=1,lmd=.03,dE=0.2,vib=z)
    x.append(z)
    y.append(1/k)
    #k = rate(Had=0.001,S=z,m=0,lmd=.03,dE=0.2,vib=0.2)
    #y2.append(1/k)

xx=np.array(x) * 1000
yy=np.array(y) * 1e12
#yy2 = np.array(y2) * 1e12

plt.rcParams.update({'font.size': 20})

fig, ax = plt.subplots(1,1, figsize = (10,6))
ax.plot(xx,yy)
#x`ax.plot(xx,yy2)
ax.set_ylim([0, 2000])

ax.set(xlabel='Odorant mode, meV', ylabel='Electron transfer time, ps',title='')

ax.grid()
```

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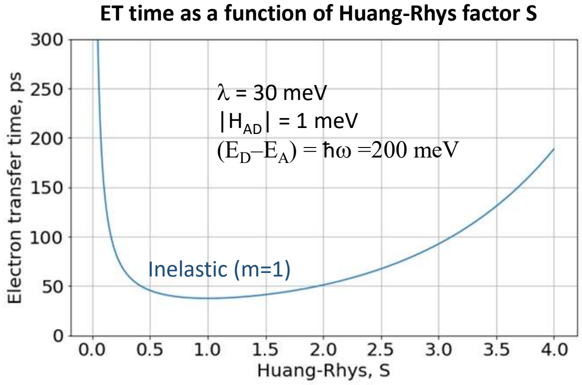
Estimating vibrational effect magnitude

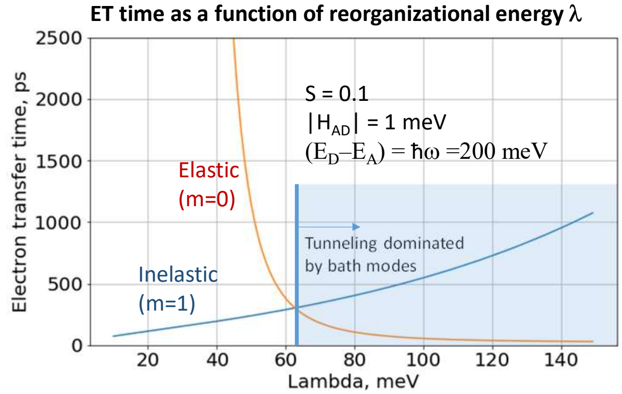
ET Transfer modeling: $k_{ET,m} = \frac{1}{\tau_{ET,m}} = \frac{2\pi}{\hbar} \left| H_{AD(el)} \right|^2 \frac{\sigma_m}{\sqrt{4\pi\lambda_{bath} kT}} \exp\left(-\frac{\left[\lambda_{bath} + \Delta E_{DA} + m\hbar\omega \right]^2}{4\lambda_{bath} kT} \right)$ $\sigma_m = \frac{e^{-S} S^m}{m!}$

ET time as a function of Huang-Rhys factor S



→ Need large S >0.1

ET time as a function of reorganizational energy λ



→ Need λ < 60 meV, ideally <40 meV

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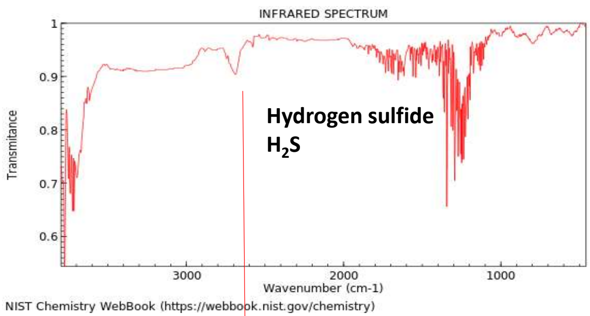
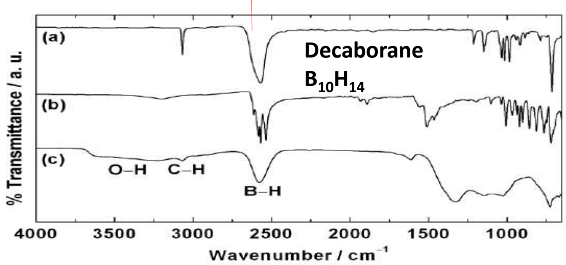
Additional evidence





Hydrogen sulfide: H_2S, vibrational mode at ~2600 cm^{-1}
Smells "sulfurous"

Similar smell:
Decaborane: $B_{10}H_{14}$
m-, o-, p- carborane: $C_2H_2B_{10}$
Also have vibrational mode at ~2600 cm^{-1} !

Note: the smell intensity correlates with IR couplings (i.e., coupling to the electric field).

https://www.researchgate.net/figure/Figure-S1-FT-IR-spectra-of-a-orthocarborane-b-decaborane-and-c-intermediate-X-as_fig1_237065828

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Molecular shape sensitivity

Some enantiomer pairs of odorants smell differently

Since odorants need to stick to protein, some shape dependence is expected.

Electron transfer rates can be affected by the shape of the odorant – it may cause changes in protein structure upon binding

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EXTRA

graphs from book and programs I wrote to calculate them…

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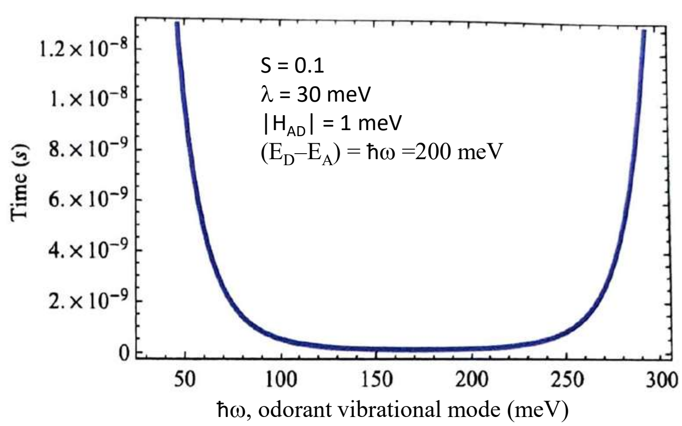
Proposed mechanisms of olfaction

ET Transfer modeling: $k_{ET,m} = \frac{1}{\tau_{ET,m}} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \frac{\sigma_m}{\sqrt{4\pi\lambda_{bath}kT}} \exp\left(-\frac{\left[\lambda_{bath} + \Delta E_{DA} + m\hbar\omega\right]^2}{4\lambda_{bath}kT}\right)$ $\sigma_m = \frac{e^{-S}S^m}{m!}$

Sensitivity to odorant mode



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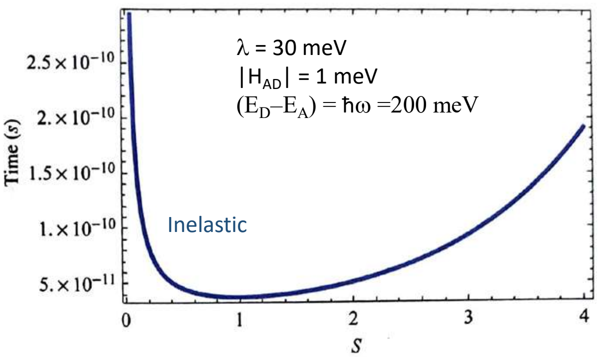
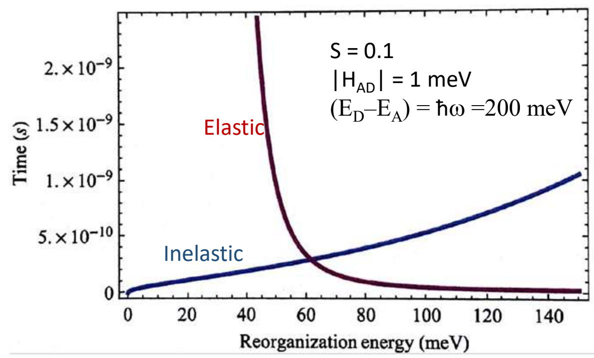
Proposed mechanisms of olfaction

ET Transfer modeling: $$k_{ET,m} = \frac{1}{\tau_{ET,m}} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \frac{\sigma_m}{\sqrt{4\pi\lambda_{bath}kT}}\exp\left(-\frac{\left[\lambda_{bath}+\Delta E_{DA}+m\hbar\omega\right]^2}{4\lambda_{bath}kT}\right)$$

ET time as a function of Huang-Rhys factor S



ET time as a function of reorganizational energy λ



PhD Thesis of J.C. Brookes

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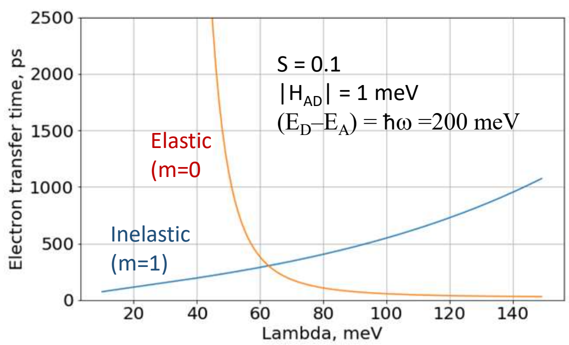
EXTRA

ET time as a function of reorganizational energy λ



Python program in notes…

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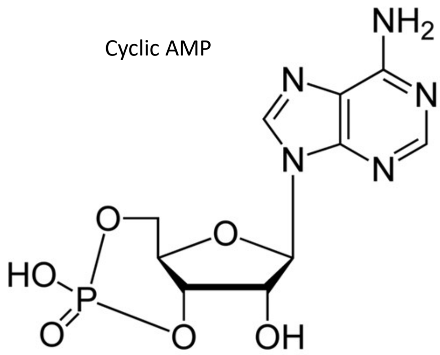
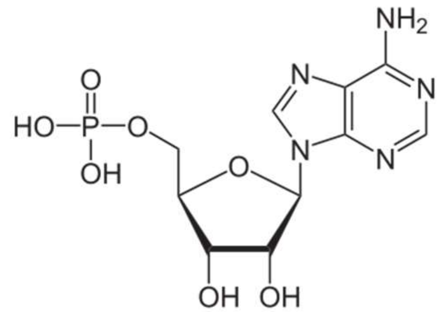
EXTRA

Cyclic AMP



AMP



Cyclic adenosine monophosphate (cAMP) is a second messenger, a derivative of ATP that is important in many biological processes such as intracellular signal transduction, etc. The most important role of cAMP is the regulation of metabolism. This could be further explained where cAMP performs a major role in the context of facilitation and mobilization promotion of glucose and fatty acid reserves.

https://www.differencebetween.com/difference-between-cyclic-amp-and-vs-amp

S. Savikhin, Purdue University

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